SANGAMITHRA R

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CAREER OBJECTIVE

Aspiring Embedded Systems Engineer with a strong foundation in C, C++, Data Structures, Microcontroller Programming, Linux Internals, and Embedded Systems. Passionate about developing efficient and reliable software solutions for embedded applications, software-hardware integration, and system optimization to solve real-world engineering challenges. Seeking an opportunity to contribute to innovative projects while continuously enhancing my technical expertise and problem-solving skills.

EXPERIENCE

Embedded Systems Trainee - Emertxe Information Technologies **February, 2026 – Present**

• Currently undergoing Advanced Embedded Systems training covering Advanced C, Data Structures, Embedded C, Linux Internals, and Microcontroller Programming.
• Gaining hands-on experience in software development using C by implementing data structures, file handling, command-line applications, and system-level programming concepts.
• Building practical projects in Advanced C and Embedded Systems while strengthening problem-solving, debugging, and code optimization skills.

EDUCATION

Bachelor of Engineering | Electronics and Communications Engineering
Saranathan College Of Engineering
- 2022 - 2026 | CGPA: 7.94

Higher Secondary School
Kamala Niketan Montessori CBSE School
- Year of Completion: 2022 | Percentage: 81%

Secondary Education
Kamala Niketan Montessori CBSE School
- Year of Completion: 2020 | Percentage: 88%

SKILLS

Programming Languages :
- C, C++, Data Structures

Embedded Systems :
- Microcontroller Programming
- GPIO, Timers, Counters, Interrupts
- UART, SPI, I2C Communication Protocols
- Software-Hardware Integration

Linux & System Programming

Technical research and Report writing

Soft Skills
- Problem Solving & Analytical Thinking
- Effective Communication
- Team Collaboration
- Adaptability and Quick Learning

ACADEMIC PROJECT

- **Trust-Score–Driven Intelligent Intrusion Detection System -** Developed an intelligent Intrusion Detection System (IDS) that utilizes trust-score analysis to identify malicious network activities and potential security threats. The system evaluates network behavior, calculates trust scores for connected entities, and classifies suspicious activities to enhance network security and reliability.

PROJECTS

- **Image Steganography using LSB Encoding and Decoding -** Developed a C-based application to securely hide and retrieve secret text messages within BMP image files using Least Significant Bit (LSB) steganography. The application encodes the message length and content into image pixels and reconstructs the original message during decoding while preserving image quality. **Technologies Used :** C Programming, File Handling, Pointers, Bitwise Operations, Structures, Command Line Arguments, Makefile
- **MP3 Tag Reader and Editor -** Developed a command-line application in C to read, display, and modify metadata stored in MP3 files using ID3 tags. The application supports viewing and editing fields such as title, artist, album, year, genre, and comments while preserving the integrity of the audio data. **Technologies Used :** Advanced C, File I/O Operations, Structures, String Manipulation, Bitwise Operations, Command Line Arguments
- **Address Book Application -** Developed a terminal-based Address Book application in C for managing contact information including names, phone numbers, and email addresses. The application supports adding, searching, editing, deleting, and displaying contacts with persistent storage using CSV files. **Technologies Used :** Advanced C, Structures, Function Pointers, File Handling, String Manipulation, Dynamic Memory Management
- **Arbitrary Precision Calculator (APC) -** Developed an Arbitrary Precision Calculator in C capable of performing arithmetic operations on numbers exceeding the limits of built-in data types. Implemented addition, subtraction, multiplication, and division using linked lists to represent and process arbitrarily large integers with high accuracy. **Technologies Used :** Advanced C, Data Structures, Linked Lists, Dynamic Memory Allocation, Self-Referential Structures, Pointers, Function Pointers

INTERESTS

- **Bharatanatyam (10 years of training)** – Cultivated discipline, expression, and stage presence
- **Public Speaking & Emceeing** – Hosted cultural events, enhancing confidence and communication skill

ACCOMPLISHMENTS

EDII Hackathon Ideation Camp – Selected participant in the Entrepreneurship Development and Innovation Institute’s ideation program focused on innovation and entrepreneurship.
Smart India Hackathon (SIH) 2024 & 2025 – Hardware Edition – Selected as one of the top 20 teams in college for two consecutive years through internal hackathon rounds.
Research on Submarine Cable Networks – Contributed to a project on AI-based failure prevention and dynamic rerouting in submarine cable systems to improve global communication reliability.